

SHP ASSET MANAGER

LAZER EYE

User Manual – V2 Hardware



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SYSTEM OVERVIEW

The **Lazer Eye** is a LiDAR-based camera trigger. It continuously measures the distance to whatever is in front of it and fires your camera shutter the moment something enters a distance window you define – no button press required.

Common uses include wildlife photography (animal crosses the beam), action sports (athlete reaches a point), product photography, and long-exposure light painting.

WIRELESS RECEIVER OPTIONS

An optional wireless receiver sits at the camera end and receives trigger and wake signals from the Lazer Eye, eliminating a cable run between the sensor and your camera. Two V2 receiver types are supported:

RECEIVER		BEST FOR
LR Mini RX V2	v2	Compact receiver. Identifies itself to the app automatically once paired.
LR Full ESP32 RX V2	v2	Full-size receiver, same feature set as the Mini.

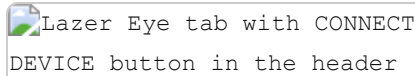
TRIGGER OUTPUT

The Lazer Eye has a wired trigger jack, and can optionally also trigger a paired V2 receiver wirelessly over ESP-NOW.

OUTPUT	WHAT IT DOES	BEST FOR
ESP-NOW wireless (optional)	Sends trigger and wake signals to a paired V2 receiver over 2.4 GHz. No cable required between the Lazer Eye and the camera.	Remote or awkward placements, water environments, moving rigs.

CONNECTING TO THE APP

The SHP Asset Manager app controls all Lazer Eye settings in real time. Connect via **Bluetooth**, **WiFi**, or through a **Pi TX** Raspberry Pi – all three give full control.



Lazer Eye tab with **CONNECT DEVICE** button in the header

Tap ⚡ **CONNECT DEVICE** to start pairing

BLUETOOTH

- 1 Power on the Lazer Eye.
- 2 Ensure Bluetooth is enabled on your phone.
- 3 Open the app and tap the **LAZER EYE** tab at the bottom of the screen.
- 4 Tap ⚡ **CONNECT DEVICE** in the header and select your device from the scan list. The Lazer Eye appears as **LASER_EYE_V2**.
- 5 Settings sync automatically. The dot in the header turns orange when connected.

The app remembers your device. On subsequent launches it reconnects silently if the Lazer Eye is in range.

WIFI

The Lazer Eye creates its own WiFi hotspot. WiFi is recommended when installing app or firmware updates – it is significantly faster than Bluetooth.

- 1 Power on the Lazer Eye.
- 2 On your **LASER_EYE** network in OS WiFi settings. If you have set a Security PIN of 8 or more characters in the app, that PIN is also the join the hotspot password.
- 3 Return to the app. Tap ⚡ **CONNECT DEVICE** → *Connect via WiFi*.
- 4 Enter IP **192.168.4.1** and port **80**, then tap **CONNECT**.
- 5 Settings sync automatically once connected.

PI TX **PI TX**

A **Pi TX** is a Raspberry Pi running the SHP Pi daemon. Its secured hotspot is the recommended field-control path: join the WiFi network whose name begins with **PI_TX_**,

then connect the app to `10.42.0.1`. No internet or separate router is required for local control. Optional Ethernet or a second WiFi adapter can provide upstream internet while the Pi's built-in WiFi remains dedicated to the hotspot.

- 1 Join the Pi TX network whose SSID `PI_TX_`, using the password printed during installation.
- 2 In the app, tap ⚡ **CONNECT DEVICE** → *Connect via WiFi*.
- 3 Choose the Pi TX connection flow. The app **10.42.0.1** first; tap CONNECT if prompted.
- 4 The app identifies the Pi `PLATFORM,PI` response and all Pi TX sections appear via its automatically.

If the Pi is also on a trusted upstream LAN, `pi-tx.local` or its LAN IP can be used instead. Normal-LAN control is currently unauthenticated, so use the secured Pi TX hotspot for untrusted or shared networks.

CONFIGURATION SETTINGS

The **CONFIGURATION** card is expanded by default at the top of the Lazer Eye tab. It holds the core trigger-zone settings, laser alignment, the LED toggles, and Wake Trigger.



Lazer Eye tab showing the configuration card with Min Distance, Max Distance, Shot Interval and Shots Per Trigger

Configuration card – Min/Max Distance, Shot Interval, Shots Per Trigger

MIN DISTANCE

0 cm up to Max Distance

The closest range at which the Lazer Eye will trigger. Objects nearer than this are ignored entirely.

MAX DISTANCE

Min Distance up to 1000 cm

The furthest range that triggers a shot. Objects beyond this are ignored. The orange zone on the distance bar shows your active trigger window.

SHOT INTERVAL

0.0 s to 30 s

Minimum time between successive triggers. Set to **0.0 s** to fire on every sensor reading. Increase to limit how rapidly the camera fires when a subject lingers in the zone.

DELAY BEFORE SHOT

0 s to 5 s

Pauses this long between the detection event and the shutter firing. Use for subjects that are still approaching the ideal frame position when detected. 0 = instant.

SHOTS PER TRIGGER

1 to 10

How many photos are taken each time the trigger fires. **1** for single capture; increase for burst sequences.

LASER ALIGNMENT

On / off – auto-off after 5 min

Fires the built-in alignment laser so you can see exactly where the sensor is aimed. Turns off automatically after 5 minutes, or tap the toggle again to stop it early.

STATUS LED

On / off

Blinks during each camera trigger and wake pulse. Disable in environments where stray light would disturb the shot.

POWER LED

On / off

Solid indicator that the Lazer Eye is powered on. Disable for stealth shooting or to minimise light pollution during long exposures.

All values are written to the device immediately when changed and are retained after power-off.

Do not stare directly into the alignment laser beam.

READING THE DISTANCE MONITOR

- > The large number at the top shows the live distance reading.
- > The **orange zone** on the bar marks your Min-Max trigger window. When the moving dot enters it the shutter fires and the display flashes **TRIGGER**.
- > Tap **CM/M** or **FT/IN** to switch the display unit.

WAKE TRIGGER

Many cameras enter a sleep or standby mode after a period of inactivity and miss the first shutter trigger after a long wait. Wake Trigger sends a short periodic pulse to keep your camera awake between triggers.

- 1 Scroll to the **WAKE TRIGGER** row inside the Configuration card.
- 2 Toggle the switch **ON** . The label turns orange when active.
- 3 Select a preset interval or enter a custom value.

PRESET	INTERVAL	RECOMMENDED FOR
30s	30 seconds	Cameras with very aggressive sleep settings
1 min	60 seconds	General use – most cameras
2 min	120 seconds	Cameras with 1-2 min auto-off
5 min	300 seconds	Long waits – wildlife setups
10 min	600 seconds	Very long waits
30 min	1800 seconds	Overnight setups
CUSTOM	1-1800 s	Any specific interval

When connected, a **NEXT PULSE IN** countdown appears at the bottom of the card showing when the next wake pulse will fire.

Set the wake interval slightly shorter than your camera's auto-off time. If the camera sleeps after 1 minute, use a 45-second interval.

TRIGGER SCHEDULE

The **TRIGGER SCHEDULE** card lets you restrict LiDAR triggering to a set time window. The shutter only fires when the current time falls inside your **START TIME**-**END TIME** range; detections outside that window are silently ignored.

SETTING A TIME WINDOW

- 1 Find the **TRIGGER SCHEDULE** card in the Lazer Eye tab and expand it.
- 2 Toggle **RESTRICT TO TIME WINDOW** on. The card label turns orange.
- 3 Enter **START** and **END** in 24 h **HH:MM**). Both fields accept a decimal keyboard –
TIME **TIME** format tap the field, type, then tap away to
(confirm.

OVERNIGHT WINDOWS

If your End Time is earlier than your Start Time the window is treated as crossing midnight. For example, **22:00**–**06:00** means “active from 10 pm until 6 am.”

LIVE STATUS BADGE

While the schedule is enabled a status banner appears below the time inputs:

BANNER	MEANING
Active – stops at HH:MM	Inside the window. LiDAR triggering is live.
Inactive – starts at HH:MM	Outside the window. Triggers are suppressed until the start time.

Times are based on the phone's local clock (24 h). Connect to the device at least once before leaving it unattended so it receives a time sync from the app.

BATTERY MONITOR

Battery Monitor watches the Lazer Eye's battery level in the background and sends a local notification when it drops below your chosen threshold – useful during long unattended shoots.

ENABLING THE MONITOR

- 1 Tap the  **gear icon** in the Lazer Eye tab header to open Settings.
- 2 Find the **BATTERY MONITOR** row and toggle it on.
- 3 Set the **ALERT** slider to the battery percentage at which you want to be notified. the **THRESHOLD** The range is 5-50%; the default is 20%.
- 4 If the system asks for notification permission, grant it. The monitor cannot alert you without permission.

When the monitor is active a small **MONITOR** badge appears in the app header next to the battery icon.

ALERT BEHAVIOUR

- > One alert fires per discharge cycle. Once the alert has fired, it will not repeat until the battery has been charged back above the threshold and drained again.
- > On **iOS** the app uses background fetch to check the battery level periodically (typically every 15 minutes). iOS schedules background fetch adaptively – the alert may arrive up to 15 minutes after the threshold is crossed.
- > On **Android** the monitor runs as a persistent foreground service with a status-bar notification so it can check the level continuously.

The monitor works over both Bluetooth and WiFi. It automatically uses silent reconnect after a brief disconnection so it does not interrupt your shoot.

ESP-NOW WIRELESS RECEIVER & PAIRING

The ESP-NOW WIRELESS RECEIVER card lets the Lazer Eye trigger a paired V2 receiver wirelessly, sending trigger and wake signals over 2.4 GHz instead of a cable run to the camera.

PAIRING A RECEIVER

- 1 Power on the wireless receiver and connect the Lazer Eye to the app (Bluetooth or WiFi).
- 2 In the Lazer Eye tab, find **ESP-NOW WIRELESS** card and turn its **ON** .
the **RECEIVER** toggle
- 3 Expand the card.  button **RECEIVER MAC** to read back the Lazer Eye's
Tap the **refresh** next to **ADDRESS** own wireless ID.
- 4 Find the MAC address printed on the receiver itself.
- 5 Enter it **PAIR** field in the

AA:BB:CC:DD:EE:FF

 , then **PAIR** .
into the **TRANSMITTER** format tap **TRANSMITTER**
MAC
- 6 The receiver locks onto the first packet it gets and pairs **NOT** to **PAIRED** .
automatically. The badge at the top of the card switches from **PAIRED**

Pairing is remembered by both devices. You do not need to re-pair after power-cycling either the Lazer Eye or the receiver.

Up to four receivers can be paired to the same transmitter simultaneously. Tap **ADD RECEIVER** to register additional units (up to the 4-receiver limit).

WHAT APPEARS AFTER PAIRING

- > **RECEIVER TYPE** — shows MINI RX V2 or FULL ESP32 RX V2 once the receiver identifies itself.
- > **RX FIRMWARE** — the receiver's firmware version, once reported.
- > **TX battery icon** — shows the receiver's battery level, updated whenever it reports a reading wirelessly.

RESETTING A RECEIVER'S PAIRING

To pair a receiver to a different Lazer Eye, clear its stored pairing first:

- 1 Power off the receiver.
- 2 Hold **BOOT** while applying power, and keep **3**, then its **button** holding for **seconds** release.
- 3 The receiver clears its stored pairing and waits for a new first packet from whichever Lazer Eye pairs to it next.

UPDATING BRIDGE FIRMWARE FROM THE APP PI TX

When connected to a Pi TX, the **BRIDGE FIRMWARE** card appears in the Lazer Eye tab. It lets you update the ESP-NOW bridge co-processor on the Pi wirelessly – no USB cable or terminal access required.

- 1 With the Pi TX connected, scroll to the **BRIDGE FIRMWARE** card.
- 2 Tap **CHECK FOR UPDATES**. The app fetches the latest .bin from the firmware GitHub release.
- 3 If an update **INSTALL** is available **UPDATE** tap OTA sends an OTA command to the Pi, which generates a fresh per-session token and forwards it to the bridge co-processor.
- 4 The bridge tears down ESP-NOW, raises a temporary WiFi access point (LR-BRIDGE-OTA), and the app joins it using the session token as the WPA2 password.
- 5 The app uploads the new firmware. The bridge reboots after flashing and re-announces itself to the Pi – the ESP-NOW card updates within a few seconds.

If no bridge co-processor is attached to the Pi, the card reports ERR_NO_BRIDGE. To abort a bridge OTA mid-transfer, power-cycle the bridge – it boots back to normal operation.

DIRECT CABLE TRIGGER CROSS-LINK

The **DIRECT CABLE TRIGGER LINK** card pairs the Lazer Eye directly with a Cable Trigger over ESP-NOW. Once paired, every accepted LiDAR detection fires the Cable Trigger wirelessly – even when your phone is put away or out of Bluetooth range.

This feature requires the Lazer Eye firmware v2.9.4 or later for confirmed-delivery status. Earlier firmware versions still send the signal but cannot confirm whether the Cable Trigger received it.

PAIRING

- 1 Connect the **Cable Trigger** to the app via Bluetooth.
- 2 Also connect the **Lazer Eye** via Bluetooth or WiFi.
- 3 In the Lazer Eye tab, find the **DIRECT CABLE TRIGGER LINK** card and expand it.
- 4 Under **STEP 1 – GET**, tap **GET CABLE TRIGGER MAC**. The app reads the Cable Trigger's ESP-NOW hardware address over BLE and fills in the MAC display automatically.
- 5 Tap **PAIR** under STEP 2. The Lazer Eye stores the Cable Trigger's MAC and begins sending wireless triggers on every detection.
- 6 The card badge switches to **PAIRED** and shows the stored MAC address.

TESTING THE LINK

Tap **TEST LINK** to send a test trigger. The result shown depends on firmware version:

- > **Delivery confirmed** – the Cable Trigger acknowledged the test packet. You should have heard a click.
- > **Sent (delivery not confirmed)** – the packet was sent but acknowledgement requires Lazer Eye firmware v2.9.4+. Listen for a click on the Cable Trigger.
- > **Error** – the packet could not be sent. Confirm ESP-NOW is enabled and the devices are within range.

UNPAIRING

Tap **UNPAIR** to remove the cross-link. Both devices forget the association. The pairing must be re-done from scratch if you want to restore it.

The cross-link is independent of your phone. Once paired you can pocket the phone – the Lazer Eye sends directly to the Cable Trigger over ESP-NOW with no relay required.

BLE CABLE TRIGGER LINK

The **CABLE TRIGGER LINK** card provides a BLE-mediated link between the Lazer Eye and the Cable Trigger. When enabled, every accepted LiDAR detection also fires the Cable Trigger via the phone's active BLE connection – no ESP-NOW pairing required, but both devices must stay connected simultaneously.

ENABLING THE LINK

- 1 Connect both **Lazer** and **Cable** to the app (either via Bluetooth or the **Eye** the **Trigger** WiFi).
- 2 In the Lazer Eye tab, find the **CABLE TRIGGER LINK** card and expand it.
- 3 Toggle **FIRE CABLE TRIGGER ON LIDAR EVENT** on. The card badge changes to **LINKED**.

A green status banner confirms both devices are connected. An amber banner appears if the Cable Trigger disconnects – the link will not fire while disconnected.

BEHAVIOUR

- > Each relay uses the Cable Trigger's own **Shots Per Trigger** setting.
- > If the Cable Trigger is already mid-sequence when a trigger arrives, that event is **skipped** – not queued. Events are not buffered.
- > The link stops automatically when either device disconnects. Toggle the switch again when both reconnect.

For hands-free setups where you want the link to work even after putting the phone away, use the Direct Cable Trigger Cross-Link (Section 08) instead. The BLE link is best when you are actively monitoring both devices from the app simultaneously.

TX & RX OUTPUT MODES

The **TX OUTPUT** card controls the Lazer Eye's own wake signal and autofocus behaviour. Once a receiver has been paired at least once, an **RX OUTPUT** card appears below it with the same controls for the receiver.

WAKE SIGNAL

Sets the signal level used on the wake/focus output.

MODE	USE WHEN
Dry Contact	Default. Works with most camera wake/focus inputs.
HOT 3.3V	Only enable if your camera requires a positive voltage pulse on the focus pin rather than a ground closure. Most cameras (Sony multi-terminal, Canon N3, Nikon MC-DC2 compatible) use ground closure – check your camera's remote/shutter pinout before switching this on.

The app shows a confirmation warning the first time you switch either Wake Signal to HOT 3.3V, since it applies voltage instead of a simple contact closure.

AUTOFOCUS PRE-PULSE

Holds the wake/focus pin for a set time *before* the shutter fires, giving your camera's autofocus a head start on moving subjects.

- > Toggle **AUTOFOCUS PRE-PULSE** (TX OUTPUT) or **RX AUTOFOCUS PRE-PULSE** (RX OUTPUT) on.
- > Set **FOCUS HOLD TIME** (TX: 50-2000 ms) or **RX FOCUS HOLD TIME** (RX: 50-5000 ms) to how long your camera needs to lock focus – typically 200-600 ms.
- > The wake pin is held for that duration, then the shutter fires.

RX OUTPUT LEDS

The RX OUTPUT card also has its own **STATUS LED** and **POWER LED** toggles for the receiver, independent of the Lazer Eye's own LED settings in the Configuration card.

RX OUTPUT is greyed out until ESP-NOW is enabled and a receiver has been paired at least once. After that it stays editable even if the Lazer Eye is disconnected.

If a red • **Command not confirmed – tap to retry** banner appears after changing a Wake Signal mode, the Lazer Eye didn't confirm the change in time and the toggle reverted. Tap the banner to resend, or move closer to the device / switch from Bluetooth to WiFi.

BROWSER PANEL

PI TX

When the app is connected to a Pi TX, the Pi also serves a full control panel to any browser connected through the Pi TX hotspot or a trusted upstream LAN. It offers live telemetry, trigger and wake controls, ESP-NOW receiver pairing, camera settings, Google Drive uploads, time-lapse, and astronomical triggers – the same feature set as the phone app.

The panel's **Camera** section mirrors the app's CAMERA card, including the three-way source selector (DSLR, Pi Camera, Webcam), the Pi Camera 3 focus and exposure controls, and the Pi Camera live preview (see Section 12). The live preview is particularly useful from a laptop – it updates at ~1 fps and lets you check framing and focus without touching the phone.

This is useful for monitoring a long shoot from a laptop, controlling the system from a second device, or accessing settings when the phone is not handy. Join the Pi TX hotspot for local browser control without internet, then open `http://10.42.0.1`.

OPENING THE PANEL

- 1 For offline field control, join `PI_TX_`. On a trusted upstream LAN, connect your computer to the same private network as the Pi instead.
- 2 Open a browser and navigate to `http://10.42.0.1` from the Pi TX hotspot, or to `http://pi-tx.local` (mDNS) from a trusted upstream LAN.
- 3 The browser panel loads automatically. No login required.

If `pi-tx.local` does not resolve, use the Pi's numeric IP address. On most home networks this is shown in your router's DHCP client list, or run `hostname -I` on the Pi.

The browser panel and the phone app use the same protocol. Both can be open simultaneously. Changes made in one are reflected in the other within a few seconds.

CAMERA

PI TX

When connected to a Pi TX, the **CAMERA** card appears in the Lazer Eye tab. It lets LiDAR triggers fire a camera attached directly to the Pi instead of the GPIO trigger pin. Three camera sources are supported, selected with the three-button source selector at the top of the card:

SOURCE	HARDWARE	BEST FOR
DSLR	DSLR or mirrorless body on the Pi's USB port (gPhoto2)	Full image quality and lens control – ISO, aperture, shutter, burst.
PI CAMERA	Raspberry Pi camera module on the CSI ribbon connector	Compact permanent installs. Pi Camera 3 adds full focus and exposure control.
WEBCAM	Any UVC USB webcam	Low-cost monitoring and documentation shots.

Only one source is active at a time. Photos from every source are saved to the Pi's staging directory and can be uploaded to Google Drive (Section 13). All settings are remembered between sessions.

DSLR / MIRRORLESS VIA USB

Any gPhoto2-compatible camera will work. This includes the majority of:

- > Canon EOS DSLRs and EOS R mirrorless bodies (PTP/PC mode)
- > Nikon DSLR and Z-series bodies
- > Sony Alpha cameras with USB PTP support
- > Fujifilm X-series bodies
- > Olympus / OM System bodies
- > Pentax DSLRs

Consult gPhoto2's camera support list at gphoto.org to verify a specific model.

- 1 Set the **PTP** (Picture Transfer Protocol) mode in its USB connection settings. This camera is usually called "PC Remote", "Tethered Shooting", or "PTP" to depending on the brand.
- 2 Connect the camera to the Pi with a USB cable.
- 3 In the app, **CAMERA** card and **DSLR** source button. Selecting the source activates expand the tap the it and starts searching for the camera.

- 4 The card switches to a searching state. Once the camera is detected, its model name and battery level appear.

Use the left and right arrow pickers to step through the available values for each setting. The Pi reads the choices directly from the camera so only values the camera supports are shown.

SETTING	DESCRIPTION
ISO	Sensor sensitivity. Choices come from the camera's own supported ISO list.
APERTURE	Lens aperture (f-number). Only shown for lenses that support electronic aperture control.
SHUTTER	Shutter speed. Choices come from the camera's own speed list including Bulb.
BURST	Number of frames captured per LiDAR trigger. Range 1-20 shots.
AUTOFOCUS DELAY	Toggle on to hold the AF signal for the Focus Hold Time before the shutter fires, giving the camera time to lock focus.

Tap **TEST SHUTTER** to fire one frame immediately and verify the connection is working.

Settings are saved to the Pi and restored automatically the next time the same camera model is connected. If the camera does not appear after plugging in, confirm it is set to PTP mode and not "mass storage" or "charging only".

PI CAMERA (CSI RIBBON)

A Raspberry Pi camera module plugged into the Pi's CSI ribbon connector can be triggered directly – no USB cable or external camera body needed. The camera must be connected before the Pi boots and enabled on the Pi (camera support is on by default in current Raspberry Pi OS; on older images enable it via `sudo raspi-config` → Interface Options).

- 1 In **CAMERA** card, **PI** source button. Selecting the source activates it and the tap **CAMERA** searches for the camera module; a green dot on the button confirms it is connected.
- 2 Once detected, the module's sensor model appears (e.g. `imx708` for a Pi Camera 3).
- 3 Choose a capture resolution with the **RESOLUTION** picker (1080p or 4K).

On camera modules older than the Pi Camera 3 (fixed-focus modules such as the v1, v2, and HQ camera with manual lenses), capture runs with automatic exposure and no focus control – the card shows the resolution picker only.

LIVE PREVIEW

Once the Pi Camera source is connected, a **live preview** appears automatically in the CAMERA card – both in the phone app and in the browser panel. It refreshes at approximately 1 frame per second and is useful for checking framing and confirming focus before a shoot or a time-lapse run.

- > The preview runs independently of captures. Taking a test shot or LiDAR-triggered shot does not interrupt it.
- > The preview stream is always 640×360 regardless of the capture resolution you have selected. It is a separate low-resolution pipeline on the Pi and has no effect on capture quality.
- > A “**Preview loading...**” message is shown while the first frame arrives – this is normal and takes a moment after switching to the Pi Camera source.

Direct browser access: the preview is also available as a still JPEG at any time by opening `http://pi-tx.local/preview.jpg` (or `http://<pi-ip>/preview.jpg`) in any browser on your network. Refreshing the page fetches a new frame. This URL works in any modern browser and is convenient for a quick focus check from a laptop without opening the full browser panel.

The preview only streams while the Pi Camera source is active. If you switch to DSLR or Webcam, `/preview.jpg` returns a 404 until you switch back to Pi Camera.

At 1 fps the preview uses very little bandwidth – typically under 50 KB/s. It can run simultaneously with a time-lapse or Drive upload without affecting those operations.

PI CAMERA 3 CONTROLS

When the connected module is a **Pi Camera 3** (Sony IMX708 sensor), the card automatically expands with focus, exposure, and white balance controls. These controls only appear on the Camera 3 – the Pi detects the sensor model at connect time.

FOCUS

CONT / AUTO / MANUAL

CONT (continuous) refocuses constantly – best for moving subjects. **AUTO** focuses once on demand using the TRIGGER AUTOFOCUS button. **MANUAL** locks the lens at a position you set.

TRIGGER AUTOFOCUS

AUTO focus mode only

Runs a one-shot focus cycle. The button shows **FOCUSING...** while the cycle runs, then **FOCUS LOCKED** or **FOCUS FAILED**. If no result arrives within about 5 seconds the button resets and you can try again. Tap again to refocus at any time.

LENS POS

EXPOSURE

MANUAL focus mode only – ∞ to ~10 cm

Steps the lens position directly. The readout shows the approximate focus distance (∞ at one end, closer distances in cm as you step up). Take a test shot to confirm focus on a fixed subject.

AUTO / MANUAL

A single mode switch for the exposure system. **AUTO** lets the camera meter both shutter and gain. **MANUAL** reveals the Shutter and Gain controls below – both are fixed together, because the sensor cannot mix an automatic shutter with a manual gain (or vice versa).

SHUTTER

1 ms to 10,000 ms

Manual exposure time in milliseconds. When you first switch Exposure to MANUAL, the current metered values are carried over as the starting point, so the image brightness does not jump.

GAIN

1× to 16× (≈ ISO 100–1600)

Manual analogue sensor gain. This is not a true ISO value – multiply by 100 for the approximate ISO equivalent shown next to the stepper. Higher gain brightens the image at the cost of noise.

WHITE BAL

AUTO / presets

Auto white balance, or a fixed preset: **Daylight, Cloudy, Tungsten, Fluorescent,** or **Indoor**. Use a preset for consistent colour across a time-lapse.

All Camera 3 settings are stored on the Pi and reapplied automatically when the camera reconnects – including after a resolution change or Pi reboot.

USB WEBCAM

Any standard UVC USB webcam can be used as a capture source. No drivers or configuration are needed on the Pi – the webcam is detected automatically.

- 1 Plug the webcam into any of the Pi's USB ports.
- 2 In **CAMERA** card, **WEBCAM** source button. Selecting the source activates it the tap the and searches for the webcam.
- 3 Once detected, the webcam's model name appears.
- 4 Choose a **RESOLUTION** picker (720p or 1080p). If the webcam does not support the exact resolution, it captures at the closest mode it does support.
capture resolution with the

Webcam capture runs with the webcam's own automatic exposure. Photos are staged and uploaded exactly like DSLR and Pi Camera shots.

GOOGLE DRIVE UPLOAD

PI TX

The **GOOGLE DRIVE** card lets the Pi TX upload photos captured by any camera source (DSLR, Pi Camera, or webcam) directly to your Google Drive. Authorization uses Google's device-flow, so no browser redirect to the Pi is needed – you approve access on any device.

STEP 1 – ENTER OAUTH CREDENTIALS

The Pi needs a Google OAuth 2.0 Client ID and Secret before it can request authorization. These are free to create:

- 1 Go to **console.cloud.google.com**, create (or select) a **Google Drive API** under APIs & Services → Library. to project, and enable the **Drive API** Library.
- 2 Go to APIs & Services → **Create Credentials** → **OAuth 2.0 Client ID**. For **TVs and Limited Input devices**. Application type choose **Input devices**.
- 3 Copy the **Client ID** and **Client Secret** shown.
- 4 In the **GOOGLE DRIVE** card, paste the Client ID and Client Secret into the fields, **SAVE TO PI**. The Pi stores them securely on its SD card. expand the and tap

STEP 2 – AUTHORIZE WITH GOOGLE

- 1 Tap **AUTHORIZE WITH GOOGLE**. The Pi starts a device-flow authorization and returns a URL and a short code.
- 2 Open the URL on any phone or computer browser, sign in to your Google account, and enter the code shown in the app. Tap the COPY URL and COPY CODE buttons to copy each to the clipboard.
- 3 After approving, return to the app and tap **ALREADY APPROVED? CHECK STATUS**. The Pi confirms authorization and shows your account email.

The Pi saves the authorization token to its SD card. You only need to authorize once – the token is refreshed automatically and survives Pi reboots.

UPLOADING PHOTOS

Once authorized, the upload controls appear:

- > **FOLDER** – type a folder name and tap SET. The Pi creates this folder in your Drive root if it does not exist. Defaults to “Pi TX Photos”.

- > **UPLOAD LAST 50** – uploads the 50 most recently captured files from the Pi's staging directory.
- > **UPLOAD ALL NEW** – uploads every file in the staging directory that has not been uploaded before.

A progress bar shows **done / total** during an active upload. Any errors (network loss, Drive quota) appear below the bar.

TIME-LAPSE

PI TX

The **TIME-LAPSE** card programs the Pi TX to fire the camera at a fixed interval for a set duration, automatically – no phone interaction required once started. The time-lapse engine uses all currently configured camera settings (ISO, aperture, shutter, burst).

CONFIGURING A SEQUENCE

FIELD	DESCRIPTION
INTERVAL	Seconds between successive triggers. Minimum 1 second.
DURATION	Total run time entered as hours + minutes. The sequence ends automatically when this time elapses.

STARTING AND STOPPING

- 1 Set INTERVAL and DURATION, then tap **START TIME-LAPSE** .
- 2 The card switches to the running view, showing a **shot counter** and **remaining time** with a progress bar.
- 3 To end the sequence **STOP** . The Pi stops firing immediately and the card early, tap **TIME-LAPSE** returns to the configuration view.

The Pi keeps the time-lapse running even if the phone disconnects mid-sequence. Reconnect at any time to check progress or stop it early.

ASTRONOMICAL TRIGGERS

PI TX

The **ASTRONOMICAL TRIGGERS** card programs the Pi TX to start a time-lapse session automatically at a calculated solar or lunar event – sunrise, sunset, moonrise, moonset, golden hour, or blue hour. Triggers survive Pi reboots and run daily without any further phone interaction.

STEP 1 – SET THE PI'S LOCATION

Astronomical times are calculated for a specific latitude and longitude. Enter the coordinates manually or use the phone's GPS:

- 1 Tap **USE CURRENT LOCATION** to fill the fields from the phone's GPS automatically. Grant location permission if prompted.
- 2 Alternatively, type a latitude and longitude directly into the fields and **SET** . tap

Once a location is set, the Pi calculates today's event times and displays them in the **TODAY'S EVENTS** list below the location fields.

EVENT TYPES

EVENT	DESCRIPTION
SUNRISE	Moment the upper limb of the sun appears above the horizon.
SUNSET	Moment the upper limb of the sun disappears below the horizon.
MOONRISE	Moment the moon rises above the horizon.
MOONSET	Moment the moon sets below the horizon.
GOLDEN HOUR START	Start of the golden hour – warm, low-angle light after sunrise or before sunset.
GOLDEN HOUR END	End of the golden hour – typically when the sun is 6° above the horizon.
BLUE HOUR START	Start of the blue hour – the twilight period with cool blue ambient light.
BLUE HOUR END	End of the blue hour – when civil twilight ends and the sky darkens fully.

STEP 2 – ADD A TRIGGER

- 1 In the **ADD TRIGGER** section, tap an event button to select it.
- 2 Set **OFFSET** in minutes. *after* the event; *before* it. For example, -15 starts
Positive negative the sequence 15 minutes
values fire values fire before sunrise.
- 3 Set **DURATION** (hours + minutes) – how long the time-lapse session should run.
- 4 Set **INTERVAL** (seconds) – time between shots during the session.
- 5 Tap **ADD [EVENT] TRIGGER** . The trigger appears in the ACTIVE TRIGGERS list.

MANAGING ACTIVE TRIGGERS

- > Each entry in **ACTIVE TRIGGERS** shows the event name, offset, duration, and interval.
- > Tap the  **delete** icon on any entry to remove that trigger.
- > Tap **CLEAR ALL** to remove every scheduled trigger at once.

Astronomical triggers are stored on the Pi and run daily. You do not need to keep the app open or the phone connected. Event times are recalculated each day using the stored location.

If a moonrise or moonset event does not occur on a given day (the moon is up all day or down all day at the set latitude), the Pi skips that day's instance automatically.

ADVANCED SETTINGS

Tap the **ADVANCED** row to expand these controls. Most users can leave them at their defaults.



Full Lazer Eye tab showing the
ADVANCED row collapsed at the
bottom of the screen

ADVANCED row – tap to expand

SCAN RATE

10–500 Hz, default 100 Hz

How many distance readings per second the sensor takes. Higher values make the system more responsive to fast-moving subjects but draw more power.

PULSE SPEED

50–2000 ms, default 300 ms

How long the trigger output is held active per shot. Increase if the camera misses short triggers.

SMOOTHING

1–10

Averages recent readings before evaluating the trigger. Higher values smooth out readings over longer distances and reduce false triggers, at the cost of slightly more latency.

FLUX GATE

Off, or 5–200

Rejects low-confidence readings (weak signal returns), which filters phantom distance spikes from fast-moving subjects crossing the beam edge or reflective surfaces. Increase if you see false triggers; lower it or turn it off if the sensor stops detecting low-reflectivity subjects (e.g. dark clothing).

STATUS LEDS & BATTERY

LAZER EYE LEDS

LED	BEHAVIOUR	MEANING
Status LED	Single blink	A trigger or wake pulse just fired. Toggle in Configuration → Status LED.
Power LED	Solid on	Device is powered on and running. Toggle in Configuration → Power LED.

V2 RECEIVER LEDS

Both the LR Mini RX V2 and LR Full ESP32 RX V2 use the same LED behaviour, controlled independently from the Lazer Eye's own LEDs via the RX OUTPUT card's Status LED and Power LED toggles.

BATTERY

The Lazer Eye's battery percentage appears as a battery icon in the app header when connected. Once a receiver is paired and reporting, its battery level appears next to the **TX** battery icon in the ESP-NOW card.

Charge devices via their USB-C ports using any standard 5 V USB charger.

TROUBLESHOOTING

False triggers when nothing is in frame

Increase the **Flux Gate** value in Advanced – start at 20 and increase until false triggers stop. The sensor is catching low-confidence reflections from a background or shiny surface.

Sensor stops detecting subjects

Lower the **Flux Gate** value, or turn it off. It may be set too strict for dark or matte subjects (e.g. black clothing). Also confirm the Min-Max trigger zone covers the subject's position using the live distance monitor.

Camera misses the first shot after a long wait

Enable **Wake Trigger** at an interval slightly shorter than your camera's auto-off time. If the camera sleeps after 1 minute, use a 45-second interval.

Camera fires too late or too early relative to the subject

Adjust **Delay Before Shot** in Configuration to fine-tune the pause between detection and the shutter firing.

Camera misses short triggers

Increase **Pulse Speed** in Advanced. Some camera bodies require a longer signal to reliably register a shutter command.

Camera doesn't focus in time before the shot

Enable **Autofocus Pre-Pulse** in TX OUTPUT (or RX OUTPUT) and increase the Focus Hold Time until the camera locks focus reliably before the shutter fires.

A red "Command not confirmed" banner appears after changing a Wake Signal mode

The command was sent but the device did not confirm it in time – the toggle reverted to the last known good state. Tap the banner to resend. If it reappears repeatedly, move closer to the device or switch from Bluetooth to WiFi.

Receiver type or firmware shows blank after pairing

Power-cycle the receiver. On boot it reports its type and firmware version to the Lazer Eye; the app displays the correct values within a few seconds of the receiver coming online.

ESP-NOW shows NOT PAIRED

Make sure the receiver is powered on and the ESP-NOW toggle is on. Re-enter the receiver's MAC address in PAIR TRANSMITTER MAC and tap PAIR TRANSMITTER again. If it was previously paired to a different Lazer Eye, reset its pairing first (see Section 07).

Can't connect via Bluetooth

Toggle Bluetooth off and on and power-cycle the Lazer Eye. On iOS: Settings → Privacy → Bluetooth → confirm the SHP Asset Manager app has permission. The Lazer Eye appears as `LASER_EYE_V2` in the scan list.

Pi TX not found / pi-tx.local does not resolve

For offline control, confirm the phone is joined to the Pi TX network whose name begins with `PI_TX_` and retry `10.42.0.1`. On a trusted upstream LAN, confirm the Pi and phone are on the same private network. If mDNS is unavailable on that LAN (some mesh routers block it), use the Pi's numeric IP address instead. Find it by running `hostname -I` on the Pi, or check the router's DHCP client list.

Trigger Schedule is blocking shots unexpectedly

Check that the phone's time is correct – the schedule uses the device clock. If an overnight window is not working as expected (e.g. 22:00–06:00), confirm that the start time is later than the end time so the window is treated as crossing midnight.

Battery Monitor alert never arrives

Confirm notification permission is granted in OS Settings for the SHP Asset Manager app. On iOS, background fetch is rate-limited by the OS – alerts may arrive up to 15 minutes after the threshold is crossed. On Android, ensure the app is not being killed by battery optimisation; add it to the battery optimisation exceptions list.

Direct Cable Trigger Cross-Link shows "delivery not confirmed"

Update the Lazer Eye firmware to v2.9.4 or later (Settings → Hardware Settings → Check For Updates). Older firmware still transmits the trigger but cannot confirm receipt. Also verify the Cable Trigger is within ESP-NOW range (typically up to 50 m line-of-sight).

BLE Cable Trigger Link fires inconsistently

Both the Lazer Eye and Cable Trigger must remain connected via BLE or WiFi simultaneously. Events are not queued – if the Cable Trigger is mid-sequence when a trigger arrives, that event is skipped. For a hands-free setup that works when the phone is away, use the Direct Cable Trigger Cross-Link (Section 08) instead.

DSLR / mirrorless not detected
after plugging in

Confirm the camera is set to **PTP** mode (not Mass Storage, Charging Only, or MTP). On Canon, this is usually labelled "PC Remote"; on Nikon, "PTP"; on Sony, "PC Remote". Power-cycle the camera after switching modes. Also check the USB cable – some cables are charge-only and carry no data.

Pi Camera not detected after
selecting the PI CAMERA source

Power off the Pi and re-seat both ends of the CSI ribbon cable – the contacts must face the correct direction (check the connector markings) and the retaining clips must be fully closed. The camera is only detected at boot, so reboot after re-seating. On older Raspberry Pi OS images, also confirm the camera interface is enabled via `sudo raspi-config` → Interface Options. Verify detection on the Pi with `libcamera-hello --list-cameras`.

Pi Camera focus / exposure controls
do not appear

The extended controls only appear on a **Pi Camera 3** (IMX708 sensor) – check the model name shown after activation. Older modules (v1, v2, HQ) are fixed-focus with automatic exposure and show the resolution picker only. If you have a Camera 3 but the model shows differently, re-seat the ribbon cable and reboot the Pi.

Webcam not found after plugging in

Confirm the webcam is a standard UVC device – run `lsusb` on the Pi to check it enumerates, and `ls /dev/video*` to confirm a video device exists. Try a different USB port, and avoid unpowered USB hubs, which can brown out the webcam. Tap the WEBCAM source button again after re-plugging.

Google Drive authorization fails or
CHECK STATUS returns an error

Confirm the Google Cloud project has the **Google Drive API** enabled (not just the credentials created). Also verify the OAuth application type is **TVs and Limited Input devices** – other application types do not support the device code flow the Pi uses. If the code has expired, tap AUTHORIZE WITH GOOGLE again to start a fresh flow.

Time-lapse stops before the set
duration

Check available storage on the Pi's SD card. If the staging directory fills up the engine halts to avoid data loss. Also confirm the USB camera stays in PTP mode and does not auto-switch to mass storage after an idle period – some cameras do this when untouched for several minutes.

Astronomical trigger does not fire
at the expected time

Confirm the Pi's system clock is correct (`timedatectl` on the Pi). Astronomical event times are calculated in UTC and converted to local time using the Pi's timezone – set

it with `sudo timedatectl set-timezone <region/city>`. Also verify the location coordinates are correct; a wrong hemisphere or sign will shift all events significantly.

Bridge firmware update fails or hangs mid-transfer

If the transfer stalls, power-cycle the bridge co-processor – it boots back to normal operation automatically. Then retry from the BRIDGE FIRMWARE card. Ensure the phone stays on the `LR-BRIDGE-OTA` WiFi network for the entire upload; switching networks mid-transfer will break it.

An update is slow or failing

Switch to a WiFi connection – updates over WiFi are dramatically faster than Bluetooth. Settings → Hardware Settings → Check For Updates.

UPDATING FIRMWARE

- 1 Connect to the Lazer Eye in the app.
 - 2 Tap the  **gear icon** in the header to open Settings.
 - 3 Scroll to **Hardware Settings** and tap **CHECK FOR UPDATES** .
 - 4 If an update is available, tap **INSTALL UPDATE** . The device reboots and the app reconnects automatically.
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